



Network Analysis

AN INTRODUCTION FOR HUMANISTS

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Modelling (Recap)



Why and How



Common models:

Erdős-Rényi & Gilbert models (no clusters)

The Watts-Strogatz model (no hubs)

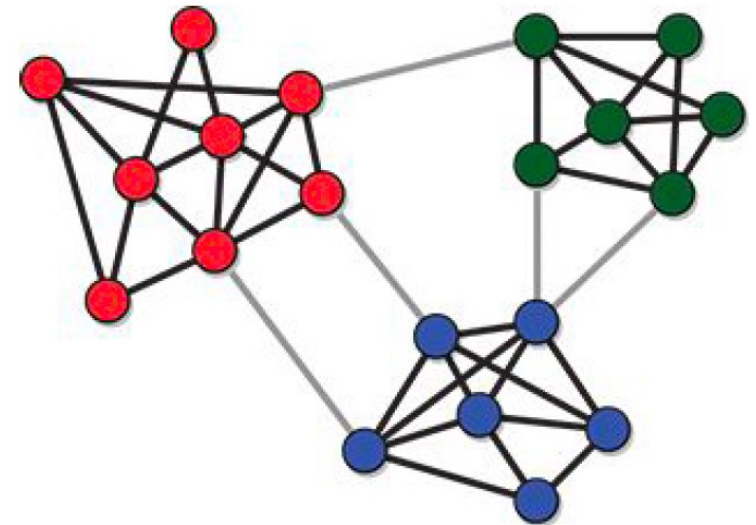
The Barabási-Albert Model (hubs, but always old)

Clusters, communities, modules

- ▶ A **community (cluster, module)** is a group of graph vertices that “belong together” according to some precisely defined criteria which can be measured.
- ▶ It is commonly defined as a group of vertices such that the **density of edges between vertices of the group is higher than the average edge density in the graph.**

- ▶ Definition based on: *Computational Complexity* ed by Meyers (2012).

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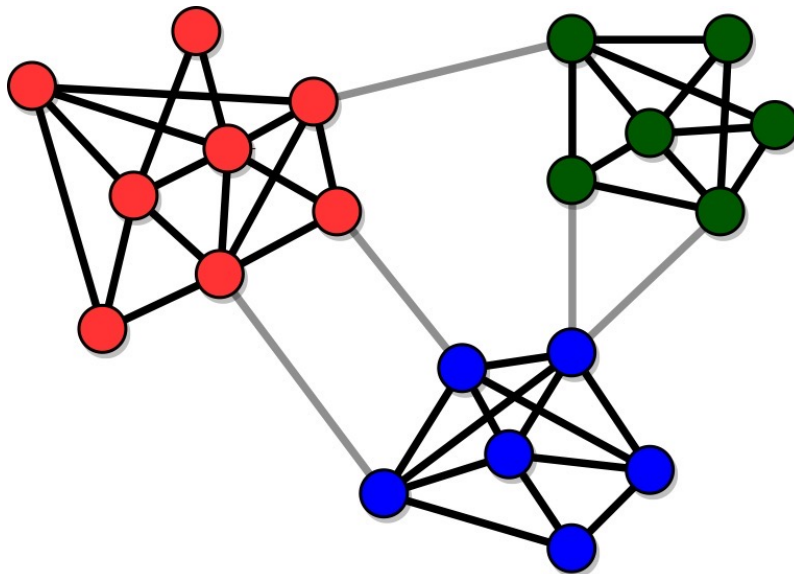


A graph with three communities indicated by node colours. Image source: Menczer, *A First Course...*, Fig. 6.1

Why study communities?

- ▶ To uncover the organization of the network
- ▶ To classify the nodes based on their position
- ▶ To uncover relationships between features of nodes and their position in the network
- ▶ To infer missing links

Basic Definitions: Communities



A graph with three communities indicated by node colours. Image source: Menczer, A First Course..., Fig. 6.1

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- **High cohesion:** communities have many internal links, so their nodes stick together
- **High separation:** communities are connected to each other by few links

Cohesion (Internal Connectivity)

- ▶ Measures how strongly connected a group of nodes is.
 - ▶ High cohesion means nodes within a group have dense intra-community links.
 - ▶ Important for community detection, resilience, and influence propagation.
- ▶ Examples:
 - ▶ Social circles in a social network.
 - ▶ Functionally similar proteins in biological networks.

Separation (External Connectivity)

- ▶ Measures how isolated a community is from the rest of the network.
 - ▶ High separation means few inter-community links.
 - ▶ Helps detect borders, bottlenecks, and inter-group influence.
- ▶ Examples:
 - ▶ Independent research domains in citation networks.
 - ▶ Different political or opinion networks.

Cohesion & Separation

▶ Communities should have **high cohesion** and **high separation**

▶ *How to measure that?*

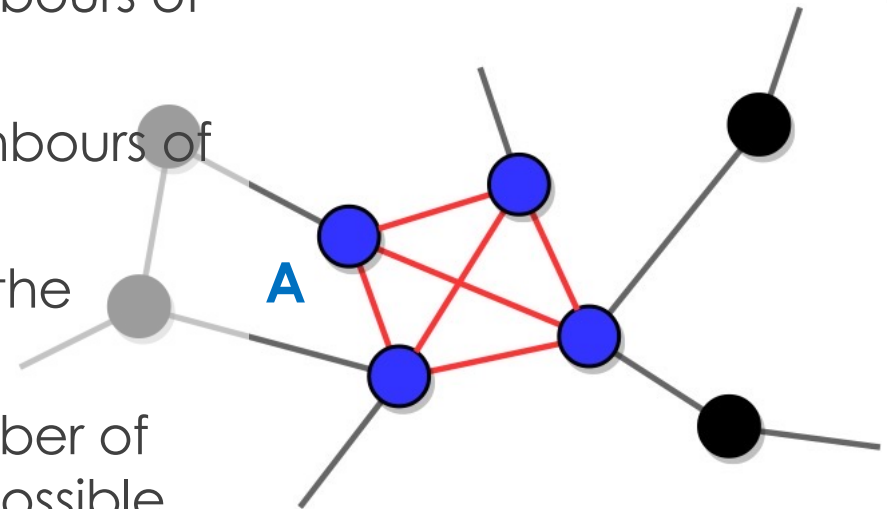


Cohesion & Separation

- ▶ Communities should have **high cohesion** and **high separation**
- ▶ *How to measure that?*
 - ▶ Check whether the number of internal links exceeds the number of external links

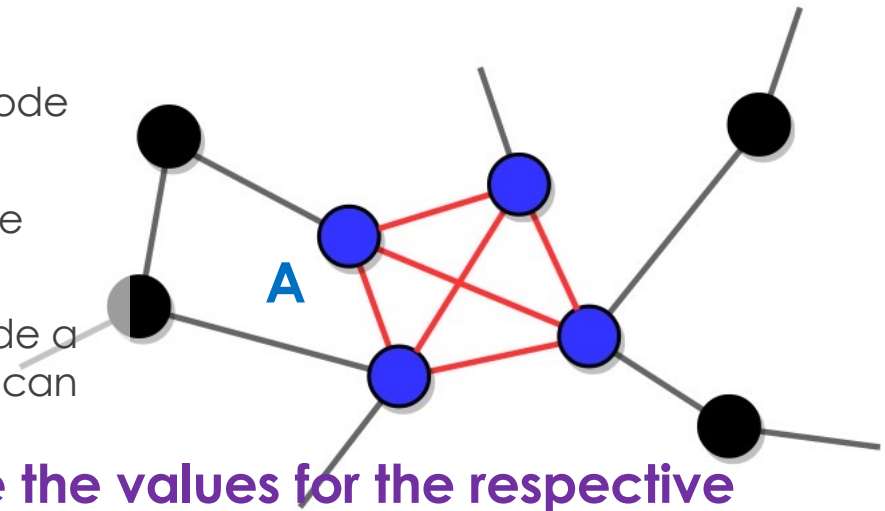
Basic Definitions: Variables in Communities

- ▶ **Internal degree of a node:** number of neighbours of the node in its community
- ▶ **External degree of a node:** number of neighbours of the node outside of its community
- ▶ **Community degree:** sum of the degrees of the nodes in the community
- ▶ **Internal link density:** ratio between the number of links inside a community and the maximal possible number of links that can lie inside this community.



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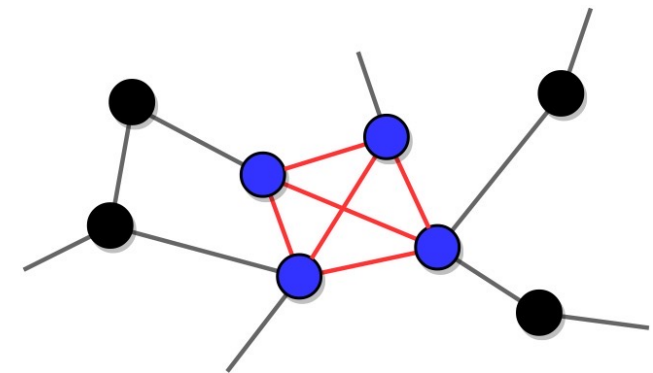


Calculate the values for the respective measures, first for node A and then for the community to which it belongs (blue nodes)

Strong & Weak Communities

- ▶ **Strong community:** subnetwork such that the internal degree of each node is greater than its external degree
- ▶ **Weak community:** subnetwork such that the sum of the internal degrees of its nodes is greater than the sum of their external degrees

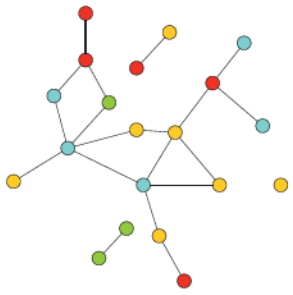
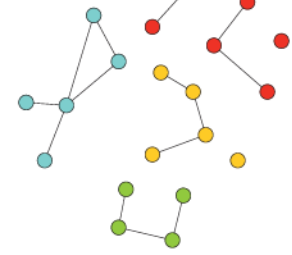
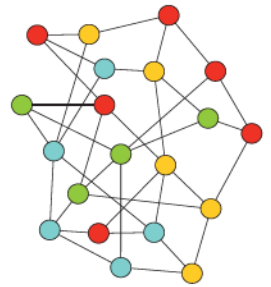
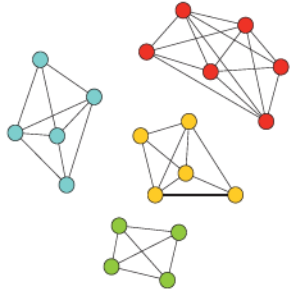
Is the blue community a strong or weak community?



Balance Between Cohesion & Separation

- ▶ Cohesion is especially important in sociology
 - ▶ Too much cohesion with little separation → overlapping clusters.
 - ▶ Too much separation with little cohesion → fragmented communities.

Nominal Group Boundary

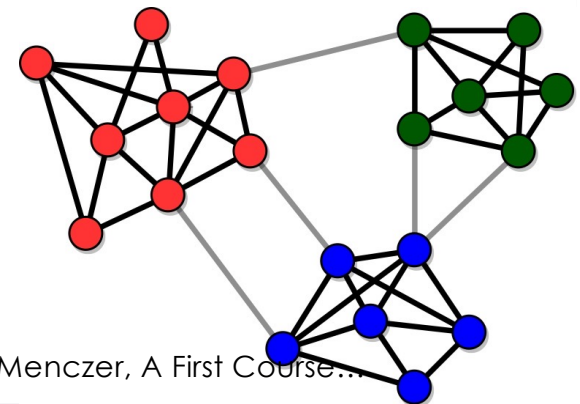
	Porous/weak	Impermeable/strong
Low	Undifferentiated, anomic social structure 	Nominal interest group structure 
High	Socially cohesive, integrated 	Segmented society, highly modular 



Grouping nodes of a network

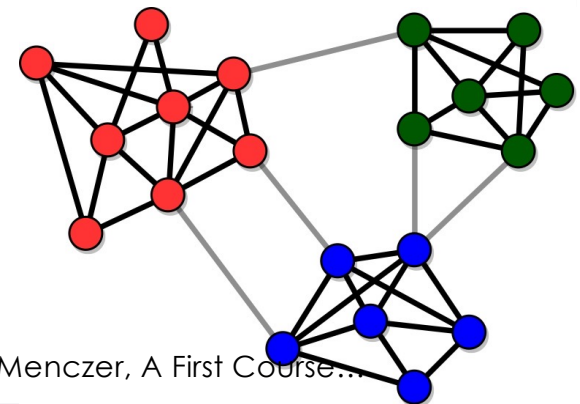
Network Partitioning

- ▶ **Partitioning** - dividing the nodes of a network into a number of groups such that the number of links between the groups is minimal
- ▶ In other words: a graph **partition** is the reduction of a graph to a smaller graph by partitioning its set of nodes into **mutually exclusive groups**.
- ▶ How many different partitions you can distinguish in the network illustrated on this slide?



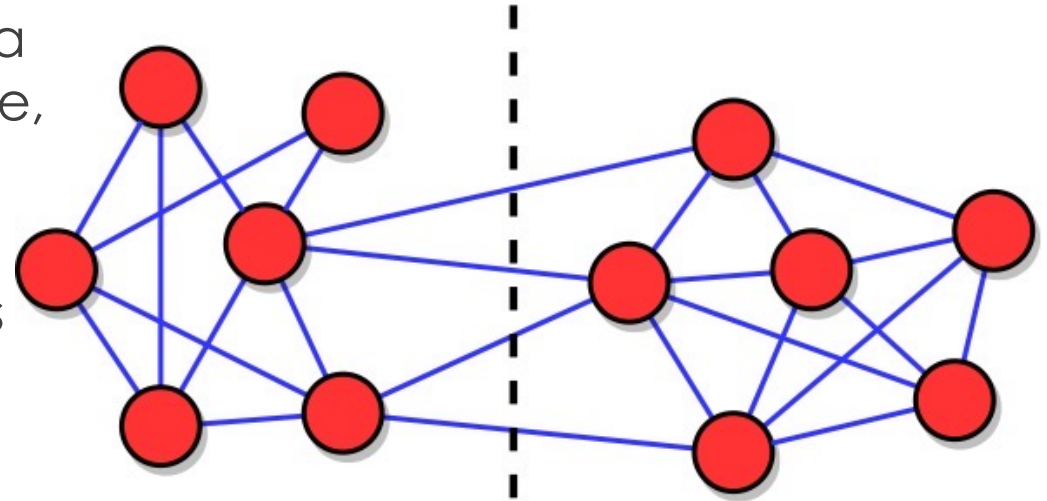
Network Partitioning

- ▶ In Network Science there are two partitions which are referred to as the **trivial partitions**:
 - ▶ 1. The partition with one set containing every node (**Singleton partition**).
 - ▶ 2. The partition with N sets, each containing a single node (**Partition of singletons**).
- ▶ A valid partition thus contains between 1 and N sets.



Network Partitioning

- ▶ **Bisectioning** - dividing the nodes of a network into two groups of equal size, such that the number of links between the groups is minimal
- ▶ The number of links between groups is called **cut size**



What is the cut size of the network on the slide?

Kernighan-Lin algorithm

- ▶ A popular algorithm used for graph bisection is **Kernighan-Lin algorithm**
- ▶ The idea behind this algorithm is simple:
 - ▶ Given an initial bisection of the network, we swap pairs of nodes between clusters, such as to obtain **the greatest decrease of the cut size**, while the size of the clusters does not change.
- ▶ **NetworkX :**
 - ▶ `kernighan_lin_bisection(G, partition=None, max_iter=10, weight='weight', seed=None)`.

Source:https://networkx.org/documentation/stable/reference/algorithms/generated/networkx.algorithms.community.kernighan_lin.kernighan_lin_bisection.html

Kernighan-Lin algorithm

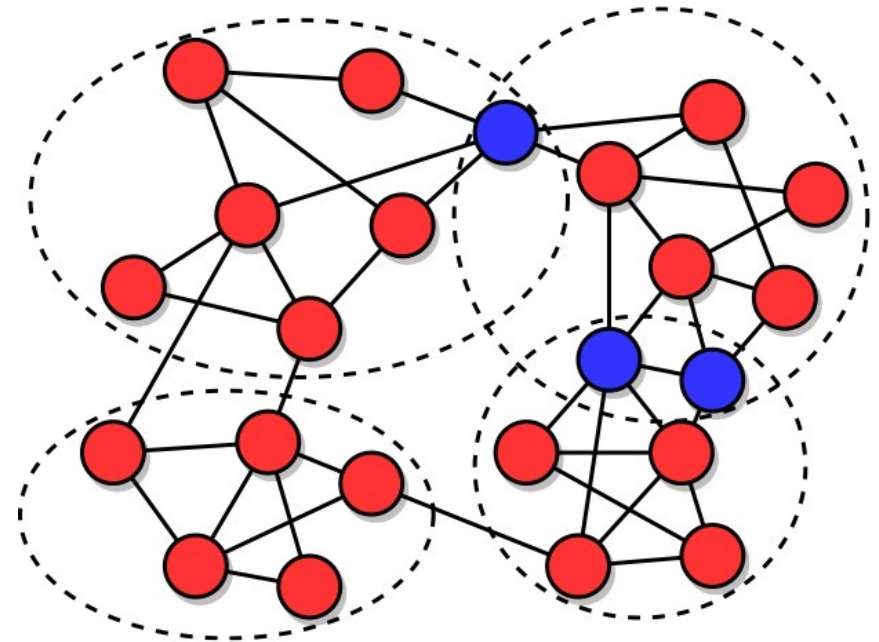
- ▶ Take a graph of N nodes and create arbitrary **Partition P** consisting of two **clusters A and B** by randomly assigning nodes to either group
- ▶ For each pair of **nodes i, j , with $i \in A$ and $j \in B$** , compute the variation in **cut size** between the current partition and the one obtained by swapping i and j .
- ▶ The pair of nodes i and j yielding the largest decrease in cut size is selected and swapped. This pair of nodes is locked; they will not be touched again during this iteration
- ▶ Repeat steps 2 and 3 until no more swaps of unlocked nodes yield a decrease in cut size. This yields a new bipartition, that is used as a starting configuration for the next iteration
- ▶ The procedure ends when the cut size of partitions obtained after consecutive iterations is the same, meaning that the algorithm is unable to improve the result.

Kernighan-Lin algorithm

- ▶ The described procedure is **greedy**, in that at each step one looks for the partition with the smallest cut size. Because of that, the algorithm gets stuck in **local optima**, i.e., solutions whose cut size is not as low as it can be.
- ▶ Greedy is an algorithmic paradigm that builds up a solution piece by piece, always choosing the next piece that offers the most obvious and immediate benefit. Greedy algorithms are used for optimization problems
 - ▶ Further tutorials on greedy algorithms here:
<https://www.geeksforgeeks.org/introduction-to-greedy-algorithm-data-structures-and-algorithm-tutorials/>

Network Partitioning vs Community Detection

- ▶ A graph **partition** is the reduction of a graph to a smaller graph by partitioning its set of nodes into **mutually exclusive groups**.
- ▶ For partitioning, clusters have to be **well-separated**, but they **do not need to have high internal link density**.
- ▶ (Usually) clusters found via graph partitioning are not communities.

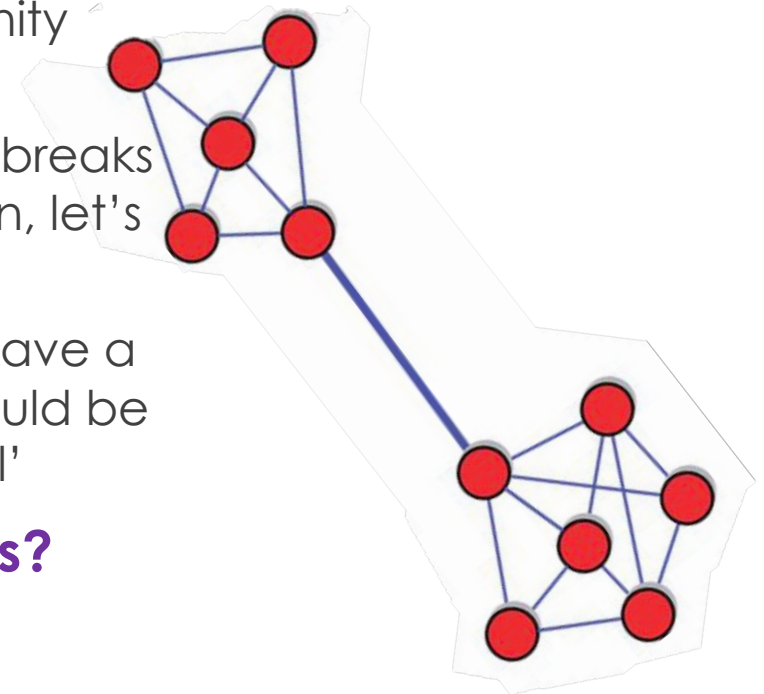


Community Detection

- ▶ Many different methods for community detection exist
 - ▶ For more info see Chapter 6 in Menczer (2020) & Chapter 11 in Newman (2010).
- ▶ Today we focus only two concepts:
 - ▶ Bridge removal
 - ▶ Modularity optimisation

Bridge Removal

- ▶ Bridge Removal is the idea behind many community detection methods.
- ▶ Strictly speaking, a **bridge** is a link whose removal breaks a connected network into two parts. For this lesson, let's call **every link joining two communities a bridge**
- ▶ 'If we were able **to locate all bridges**, we would have a natural way to detect the clusters' => clusters would be the connected components after bridge removal'
- ▶ **But how to mathematically identify bridges?**



Bridge Removal

- ▶ We can use **link betweenness** to identify bridges:
 - ▶ ‘**Bridges** are expected to have **large betweenness** values because shortest paths between nodes in different communities run through bridges’
 - ▶ ‘**Internal links** are expected to have **comparatively lower betweenness** values, because there are many alternative routes going from one node of the community to another, due to the high density of links inside the cluster’
- ▶ Link betweenness is used, for example, in **Girvan-Newman algorithm**

Girvan-Newman Algorithm: Procedure

- ▶ Start by computing the betweenness for all links
- ▶ Each iteration consists of two steps:
 - ▶ Remove the link with largest betweenness (in case of ties, one of them is picked at random)
 - ▶ Recalculate the betweenness of the remaining links
- ▶ The algorithm stops when all links are removed, so all the nodes are isolated

Girvan-Newman Algorithm: Limits

- ▶ The recalculation of the betweenness at each iteration is necessary, but it makes the algorithm slow
 - ▶ not practical for large networks with, let's say, more than 10,000 nodes.
- ▶ The method delivers a full hierarchy of N partitions, from the one consisting of a single partition covering the entire network (**Singleton partition**) to the one where each node is in its own partition (**Partition of singletons**).
- ▶ Which ones with elements between 1 and N are meaningful communities?
- ▶ **How can we select best partitions that allow us to identify communities?**



Question: How can we evaluate how good a partition is?

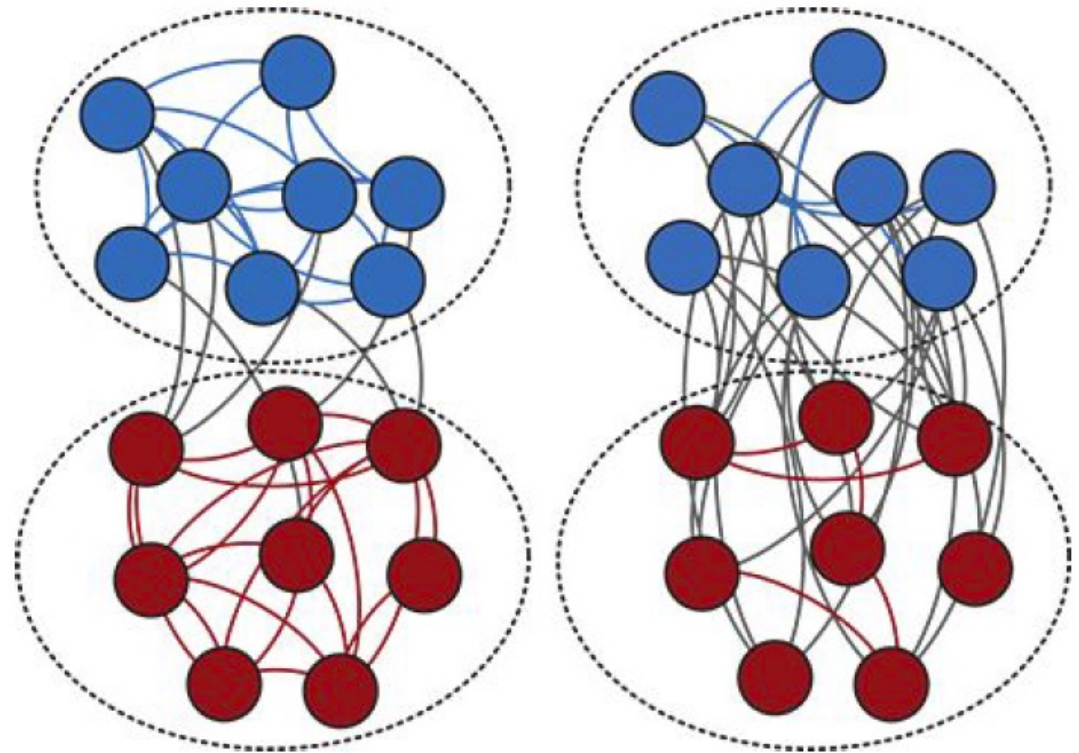


Answer: Modularity Optimisation

Modularity

- ▶ ‘The **modularity of a partition evaluates the communities** not in absolute terms, but **with respect to a random baseline**. It does so by discounting the internal links that could be attributed to a randomized version of the original network.’
- ▶ **Baseline:** randomized versions of the original network, preserving its degree sequence
- ▶ For each community of a partition, modularity computes **the difference between the number of internal links in the community and the expected value of this number in the set of randomized networks**

- If the **network is random** the **modularity** of any partition is supposed to be **low**, because the number of internal links of any cluster of the partition should be close to the expected value in the randomized networks
- If the **number of links within the clusters** is much larger than its expected random value **modularity can reach high values**



Network modularity (left – real, right – the same but randomized)

Image source: Menczer (2020), based on Fortunato and Hric (2016).

Modularity

Higher modularity (~ 0.3 to 0.7) $\rightarrow$ Stronger community structure.

Lower modularity (~ 0.0 to 0.1) $\rightarrow$ Weak or no community structure.

Negative modularity $\rightarrow$ Worse than random clustering.

Modularity

The modularity of a partition in an undirected, unweighted network is defined as

$$Q = \frac{1}{L} \sum_C \left(L_C - \frac{k_C^2}{4L} \right), \quad (6.4)$$

where the sum runs over all clusters of the partition, L_C is the number of internal links in cluster C , k_C is the total degree of the nodes in C [Eq. (6.2)], and L is the number of links in the network.

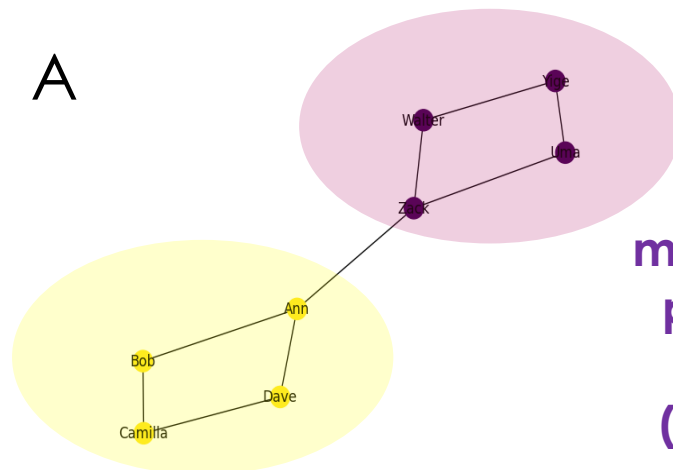
L - Total number of edges in the network
 $\sum_C$ Sum over all communities C
 L_C Number of edges *within* community C
 k_C Sum of degrees of all nodes in community C (i.e. $k_C = \sum_{i \in C} k_i$)
 $k_C^2 / 4L$ Expected number of edges within community C under the random null model

Source: Equation 6.4 in Menczer et al.

For more: Newman & Girvan (2024) Finding and evaluating community structure in networks

<https://arxiv.org/pdf/cond-mat/0308217>

A



Calculate the
modularity for these two
partitions of the same
network
(with 2 and 3 clusters)

B

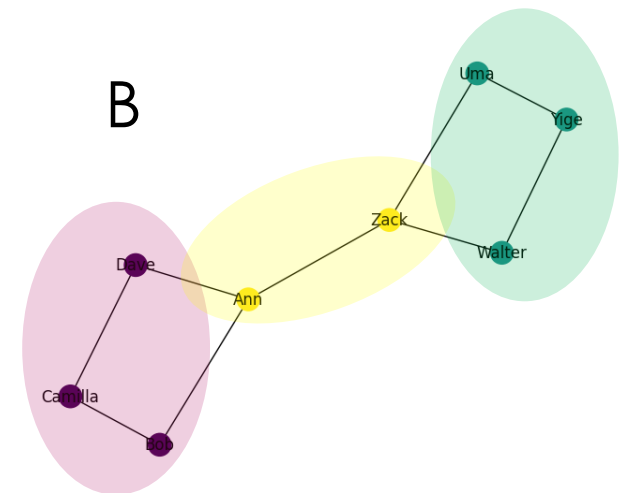


Image source: Menczer, A First Course...

Modularity

- ▶ How to find **the best partition** from a list of partitions generated by, for example, the Girvan-Newman algorithm?
 - ▶ **Modularity Maximisation:** Search for the partition with **the largest possible modularity**.
 - ▶ **How?**
 - ▶ Explore a hierarchy of partitions, with the initial partition into singletons (which has negative modularity).
 - ▶ The modularity increases steadily until a positive peak is reached.
 - ▶ Then it goes down until it hits zero when all nodes are in the same community.
 - ▶ The partition corresponding to the peak value is the best solution found by the algorithm.

NetworkX

```
greedy_modularity_communities(G, weight=None, resolution=1, cutoff  
                               =1, best_n=None)
```

This function uses the Clauset-Newman-Moore greedy modularity maximization to find the community partition with the largest modularity

Source:https://networkx.org/documentation/stable/reference/algorithms/generated/networkx.algorithms.community.modularity_max.greedy_modularity_communities.html#rce363827c0a4-2

Clauset, A., Newman, M. E., & Moore, C. 'Finding community structure in very large networks', Physical Review E 70(6), 2004.

NetworkX

```
modularity(G, communities, weight='weight')
```

- ▶ This function returns the modularity of the given partition of the graph.
- ▶ Source: <https://networkx.org/documentation/networkx-2.5/reference/algorithms/generated/networkx.algorithms.community.quality.modularity.html>